

You are a SOC Analyst at Mighty Solutions, Inc. An account executive, Dana Derringer, noticed a warning email in her inbox claiming her online access has been disabled. However, she noticed this was odd as she is still able to access her online business platforms and inbox. She decided to forward the email in question to the security team's phishing mailbox for review.

Q1 Based on the contents of the email header, what is the full date and time of the email delivery?

Windows Server 2012 R2

File Edit Format View Help

d=socialhelwanedu.onmicrosoft.com;
s=selector1-socialhelwanedu.onmicrosoft.com;
h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-SenderADCheck;
bh=EXQFetuxr4x1Qx0YHx6DbZa6FqhiXpOuxGDP5QKht+;
b=Ogub1sx+V3Z+1/V0M19NDKd9bQh0iRzB3UXgvf3ZP5HCN4S6nr4ulJhgE+K7Z9e2QdX3Zfbicr5A16GVKfA2m15SxezXtWYg7bWhZCNCKJfJHIAUG8G/PETP81an5CkgEDI37Ca+45XtAHiA64bexkU15Tn7q

Authentication-Results-Original: dkim=none (message not signed)
header.d=none;dmARC=none action=none header.from=social.helwan.edu.eg;
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b1:33) by DB8PR01066358.eurprd01.prod.exchangelabs.com
(2603:10a6:10:159:17) with Microsoft SMTP Server (version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6954.19; Tue, 31 Oct
2023 19:10:04 +0000
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80:d788:f72d:2145:caa4]) by
VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80:d788:f72d:2145:caa45]) with mapi id 15.20.6933.029; Tue, 31 Oct 2023
19:10:04 +0000
Date: Tue, 31 Oct 2023 19:10:04 -0900
To: dderringer@mighty-solutions.net
From: Outlook Support Team <social201511138@social.helwan.edu.eg>
Subject: Your account has been flagged for unusual activity
Message-ID: <JMrByPl2c3H8085ctKn75C5Gp64sP5SWk76p45jQ@s6>
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
(2603:10a6:102:347:14) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b1:33)
Return-Path: social201511138@social.helwan.edu.eg
X-Sender-IP: 40.107.22.60
Importance: high
MIME-Version: 1.0

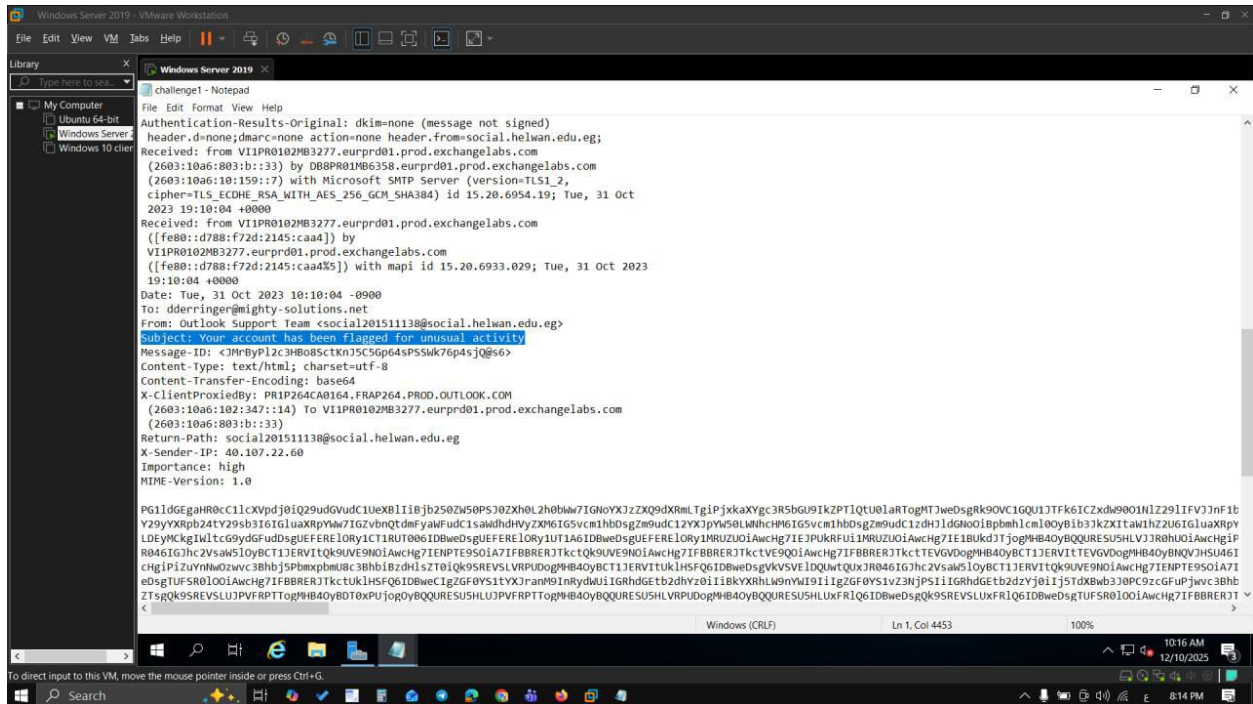
PG1dGEGaH0c1cXvpdJ0iQ29udGvudC1uExB1I8jb250Z5W0PS10Zxh0L2h0bW7IGh0YXJzZXQ9dXRmlTgPjxkaXVgc3R5bGU9KzptLTQ1a0aTogMTJweDsgrk9OVC1GQU1JTFk6TCZxdh001N1Z291IFVj1Nf1JnF1Z29yYXRk24tY295b3I6IGlUaXRxYmVIG2vbnQtdmF4YudC1saWdhbWVZXMG1G5vcmlhbsD9zm9udC12YXZpY5U0bWdhcHw1G5vcmlhbsD9zm9udC12dH1dGnoo1Bpbhclm100bY1b3kZXItaWtlZ2U6IGlUaXRxYmV1

Windows (CRLF) Ln 1, Col 4291 100%

11:26 AM 12/10/2025

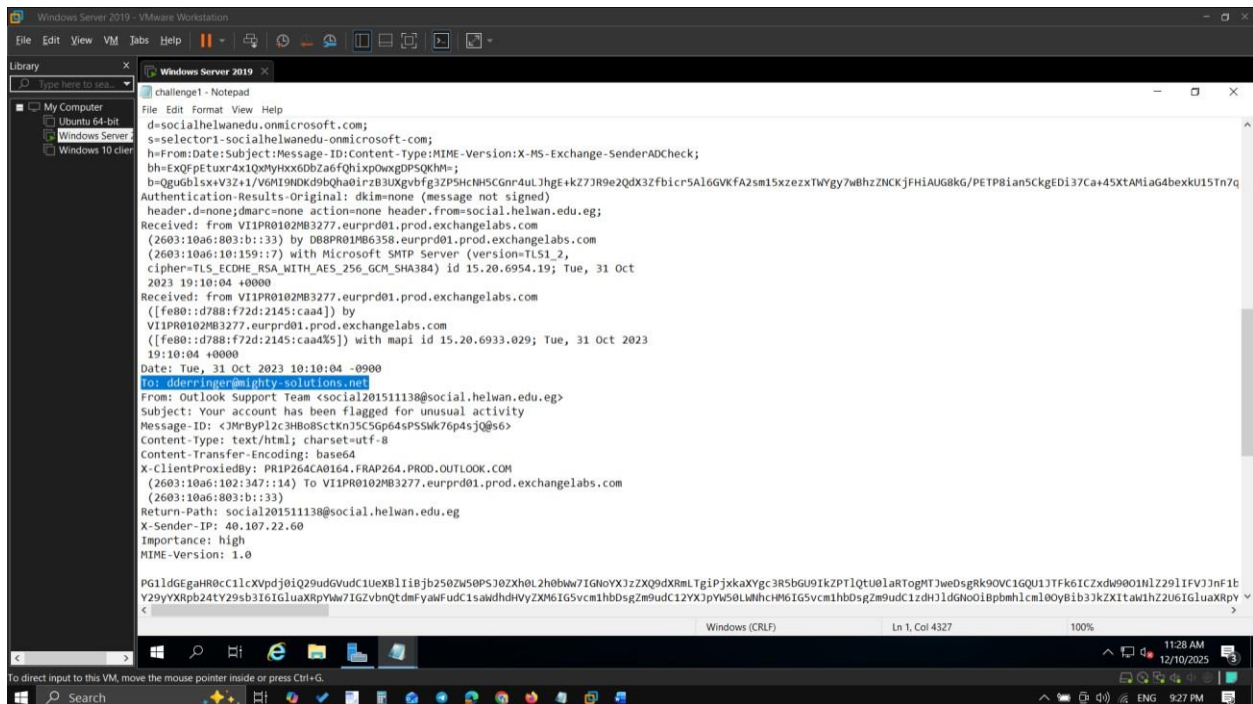
Q2 What is the subject of the email?

Your account has been flagged for unusual activity



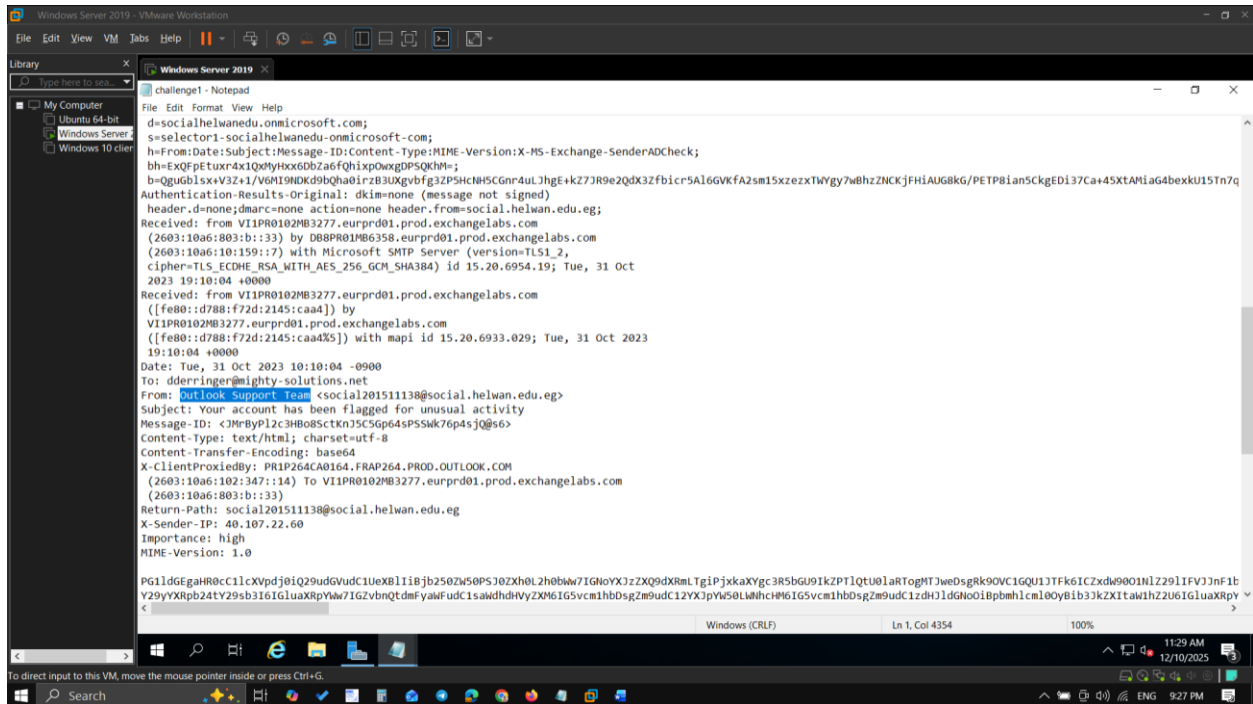
Q3 Who was the email sent to?

dderringer@mighty-solutions.net



Q4 Based on the sender's display name, who does the email claim to be from?

Outlook Support Team

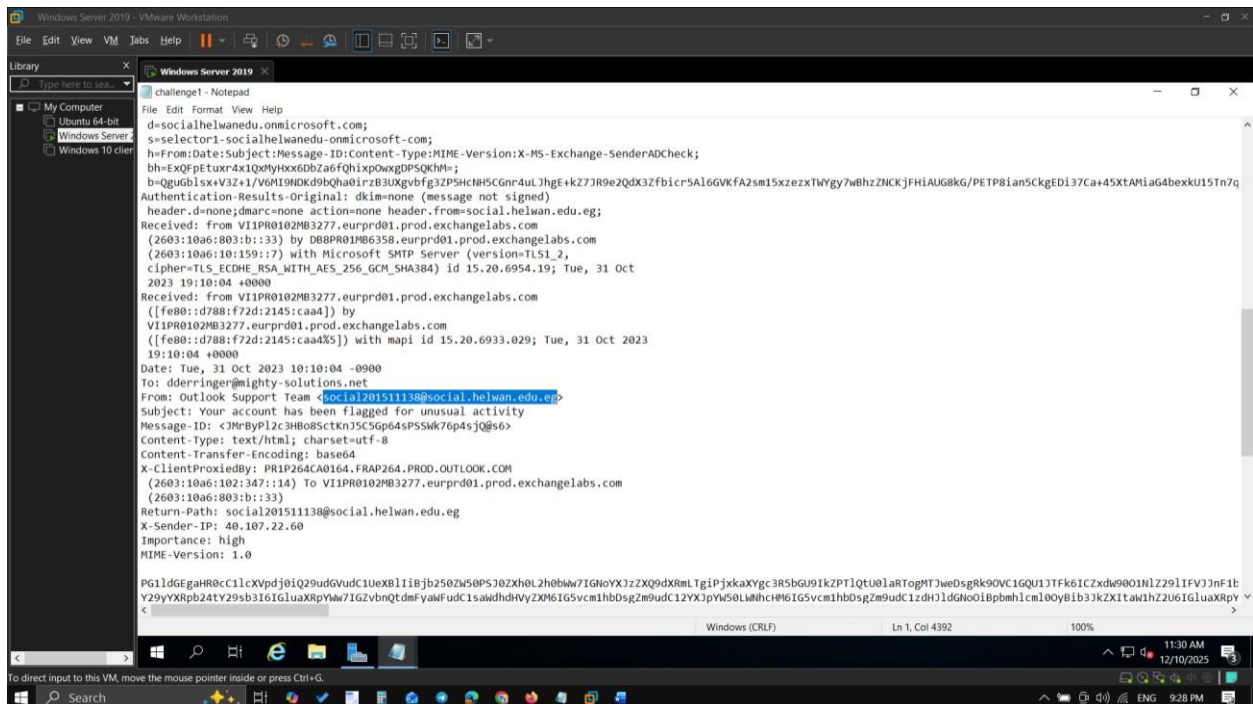


```
File Edit Format View Help
d=socialhelwanedu.onmicrosoft.com;
s=selector1-socialhelwanedu-onmicrosoft-com;
h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-SenderADCheck;
bh=ExQFpETuxr4x1QpYHx6D0Za6fQh1xP0wGDP5QKHm;
b=QuGblSxv3Z+1/V6M19NDKd9bQha01rZB3UXgvbfg3ZPSHCNH5Gcnr4uL3hgE+kZ7JR9e2QdX3Zfbicr5A16GVKFA2sm15xezzTWygy7wBhZCNCKJFHIAUG8K6/PETP8ian5CkgEDi37Ca+45XTAMiaG4bexku15Tn7q
Authentication-Results-Original: dkim=none (message not signed)
header.d=none;dmarc=none action=none header.from=social.helwan.edu.eg;
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33) by DB8PR01MB6358.eurprd01.prod.exchangelabs.com
(2603:10a6:102:159:17) with Microsoft SMTP Server (version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6954.19; Tue, 31 Oct
2023 19:10:04 +0000
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80:d788:f72d:2145:caa4]) by
VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80:d788:f72d:2145:caa435]) with mapi id 15.20.6933.029; Tue, 31 Oct 2023
19:10:04 +0000
Date: Tue, 31 Oct 2023 10:10:04 -0900
To: dderringer@mighty-solutions.net
From: Outlook Support Team <social201511138@social.helwan.edu.eg>
Subject: Your account has been flagged for unusual activity
Message-ID: <3MrByPl2c3H80Sctkn75C5Gp64SP5Skw76p45jQ@s6>
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
(2603:10a6:102:147:114) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33)
Return-Path: social201511138@social.helwan.edu.eg
X-Sender-IP: 40.107.22.60
Importance: high
MIME-Version: 1.0

Pg1ldGEgAHR0c1lcXVpdj0iQ29udGvudC1ueXB1IiBjb250Z5W0PSJ0ZXh0L2h0bWw7IGNoYXJzZXQ9dXRmLTg1PjxkaXYgc3R5bGU9IktZPT1QtU0laRTogMTJweDsgRk90VC1GQU1JTfK6ICZxdw9001NlZ291IFV3JnF1b
Y29yYXRpb24Y29sb3I6IGluaXRPYw7IGZvbntQdmFyaWZudC1sawdhdrvY2XN6IG5vcmlhbDsgZm9udC1zdHJldGNoO1Bpbmhlcm10YBib3JkZXIaw1hZ2U6IGluaXRPYw
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Q5 What is the sender's email address?

social201511138@social.helwan.edu.eg

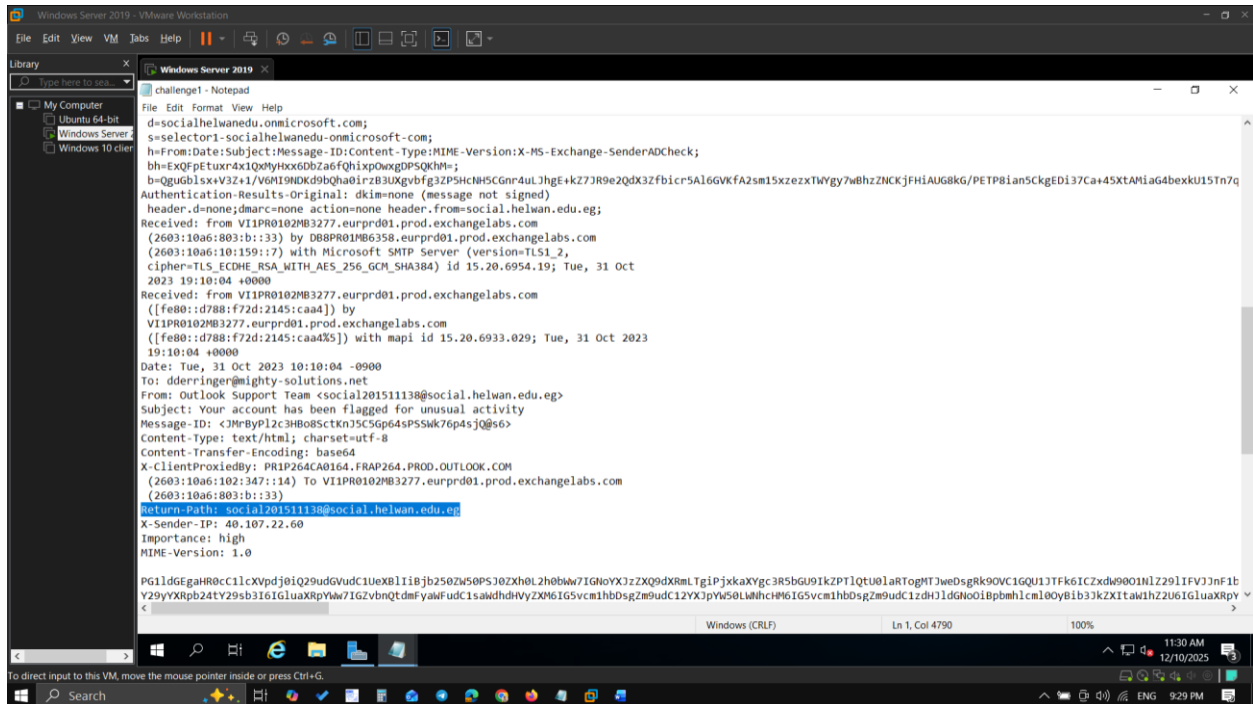


```
File Edit Format View Help
d=socialhelwanedu.onmicrosoft.com;
s=selector1-socialhelwanedu-onmicrosoft-com;
h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-SenderADCheck;
bh=ExQFpETuxr4x1QpYHx6D0Za6fQh1xP0wGDP5QKHm;
b=QuGblSxv3Z+1/V6M19NDKd9bQha01rZB3UXgvbfg3ZPSHCNH5Gcnr4uL3hgE+kZ7JR9e2QdX3Zfbicr5A16GVKFA2sm15xezzTWygy7wBhZCNCKJFHIAUG8K6/PETP8ian5CkgEDi37Ca+45XTAMiaG4bexku15Tn7q
Authentication-Results-Original: dkim=none (message not signed)
header.d=none;dmarc=none action=none header.from=social.helwan.edu.eg;
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33) by DB8PR01MB6358.eurprd01.prod.exchangelabs.com
(2603:10a6:102:159:17) with Microsoft SMTP Server (version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6954.19; Tue, 31 Oct
2023 19:10:04 +0000
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80:d788:f72d:2145:caa4]) by
VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80:d788:f72d:2145:caa435]) with mapi id 15.20.6933.029; Tue, 31 Oct 2023
19:10:04 +0000
Date: Tue, 31 Oct 2023 10:10:04 -0900
To: dderringer@mighty-solutions.net
From: Outlook Support Team <social201511138@social.helwan.edu.eg>
Subject: Your account has been flagged for unusual activity
Message-ID: <3MrByPl2c3H80Sctkn75C5Gp64SP5Skw76p45jQ@s6>
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
(2603:10a6:102:147:114) To VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33)
Return-Path: social201511138@social.helwan.edu.eg
X-Sender-IP: 40.107.22.60
Importance: high
MIME-Version: 1.0

Pg1ldGEgAHR0c1lcXVpdj0iQ29udGvudC1ueXB1IiBjb250Z5W0PSJ0ZXh0L2h0bWw7IGNoYXJzZXQ9dXRmLTg1PjxkaXYgc3R5bGU9IktZPT1QtU0laRTogMTJweDsgRk90VC1GQU1JTfK6ICZxdw9001NlZ291IFV3JnF1b
Y29yYXRpb24Y29sb3I6IGluaXRPYw7IGZvbntQdmFyaWZudC1sawdhdrvY2XN6IG5vcmlhbDsgZm9udC1zdHJldGNoO1Bpbmhlcm10YBib3JkZXIaw1hZ2U6IGluaXRPYw
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Q6 What email address is used for receiving bounced emails?

social201511138@social.helwan.edu.eg

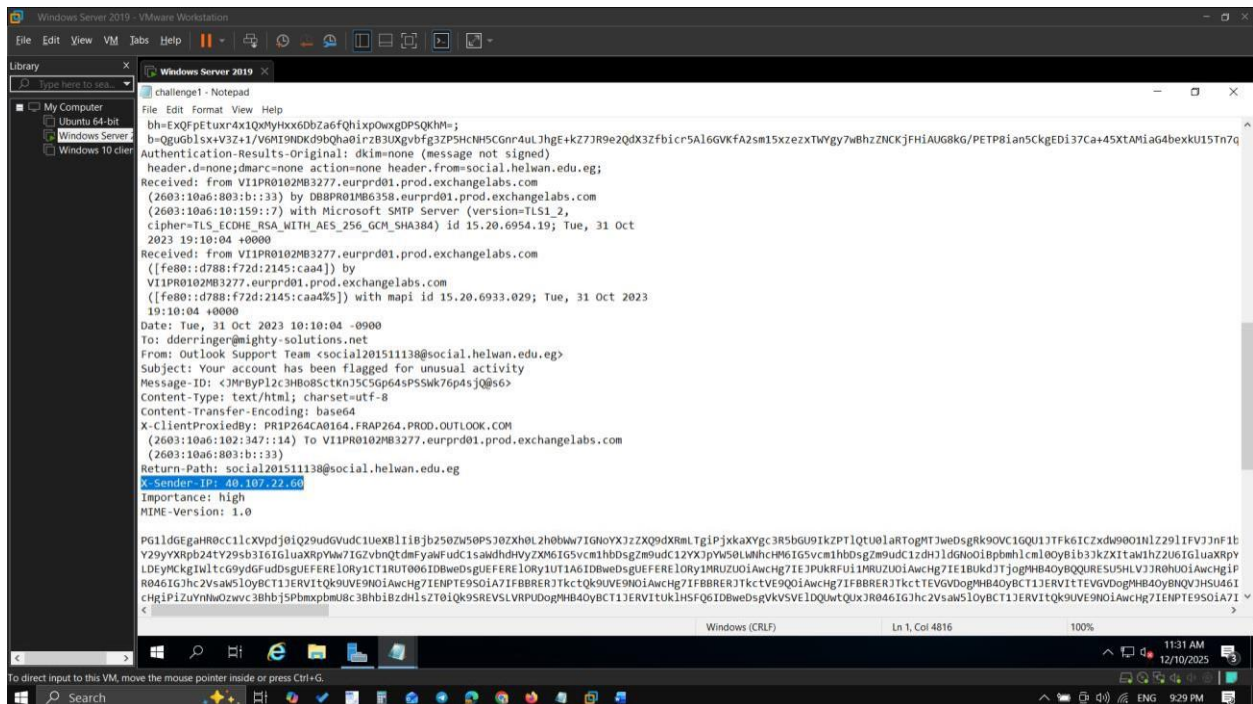


```
File Edit Format View Help
d=socialhelwanedu.onmicrosoft.com;
s=selector1-socialhelwanedu-onmicrosoft-com;
h=From:Date:Subject:Message-ID:Content-Type:MIME-Version:X-MS-Exchange-SenderADCheck;
bh=ExQfPEtuxr4x1QqyHxx6DbZa6fQhixP0xwGDP5QkHm;
b=QguGblsx+V3Z+1/V6MI9NDKd9bQha0irZB3UXgvbfG3ZPSHCNH5C6n4ulJhgE+kZ7JR9e2QdX3Zfbicr5A16GVKFA2sm15xezXTWYg7wBhzZNCKJFHIAUG8K6/PETP8ian5CkgEDi37Ca+45XTAMiA64bexku15Tn7q
Authentication-Results-Original: dkim=none (message not signed)
header.d=none;dmarc=none action=none header.from=social.helwan.edu.eg;
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33) by DB8P801MB6358.eurprd01.prod.exchangelabs.com
(2603:10a6:102:159:17) with Microsoft SMTP Server (version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6954.19; Tue, 31 Oct
2023 19:10:04 +0000
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80::d788:f72d:2145:caa4]) by
VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80::d788:f72d:2145:caa45]) with mapi id 15.20.6933.029; Tue, 31 Oct 2023
19:10:04 +0000
Date: Tue, 31 Oct 2023 10:10:04 -0900
To: dderringer@mighty-solutions.net
From: Outlook Support Team <social201511138@social.helwan.edu.eg>
Subject: Your account has been flagged for unusual activity
Message-ID: <JMrByPl2c3H8o8SctKn75C5Gp64SP5SkW76p45jQ@s>
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
(2603:10a6:102:347::14) To: VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33)
Return-Path: social201511138@social.helwan.edu.eg
X-Sender-IP: 40.107.22.60
Importance: high
MIME-Version: 1.0

Pg1ldGEgAHRCc1lcXVpdj0lQ29udGvudC1ueXB1IiBjb250Z5wPSj0ZXB0Z2h0bWw7IGNoYXJzZXQ9dXRmLTg1PjxkaXYgc3R5bGU91kZPTlQtU0laRTogMTJweDsgRk90VC1GQU1JTfK6ICZxdw9001NlZ291IFVjJnF1b
Y29yYXRpb24tY295b3I6IGluaXRPYw7IGZvbntQdmFyaWZudC1sawdhdrhvyZXN6IG5vcmlhbdsgZm9udC12YXJpYw50LWlhchM6IG5vcmlhbdsgZm9udC1zdHJldGNoO1Bpbmhlcm100yBib3JkZXItaW1hZ2U6IGluaXRPY
<
```

Q7 What is the IP address of the sender's email server?

40.107.22.60



```
File Edit Format View Help
bh=ExQfPEtuxr4x1QqyHxx6DbZa6fQhixP0xwGDP5QkHm;
b=QguGblsx+V3Z+1/V6MI9NDKd9bQha0irZB3UXgvbfG3ZPSHCNH5C6n4ulJhgE+kZ7JR9e2QdX3Zfbicr5A16GVKFA2sm15xezXTWYg7wBhzZNCKJFHIAUG8K6/PETP8ian5CkgEDi37Ca+45XTAMiA64bexku15Tn7q
Authentication-Results-Original: dkim=none (message not signed)
header.d=none;dmarc=none action=none header.from=social.helwan.edu.eg;
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33) by DB8P801MB6358.eurprd01.prod.exchangelabs.com
(2603:10a6:102:159:17) with Microsoft SMTP Server (version=TLS1_2,
cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6954.19; Tue, 31 Oct
2023 19:10:04 +0000
Received: from VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80::d788:f72d:2145:caa4]) by
VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
([fe80::d788:f72d:2145:caa45]) with mapi id 15.20.6933.029; Tue, 31 Oct 2023
19:10:04 +0000
Date: Tue, 31 Oct 2023 10:10:04 -0900
To: dderringer@mighty-solutions.net
From: Outlook Support Team <social201511138@social.helwan.edu.eg>
Subject: Your account has been flagged for unusual activity
Message-ID: <JMrByPl2c3H8o8SctKn75C5Gp64SP5SkW76p45jQ@s>
Content-Type: text/html; charset=utf-8
Content-Transfer-Encoding: base64
X-ClientProxiedBy: PR1P264CA0164.FRAP264.PROD.OUTLOOK.COM
(2603:10a6:102:347::14) To: VI1PR0102MB3277.eurprd01.prod.exchangelabs.com
(2603:10a6:803:b:33)
Return-Path: social201511138@social.helwan.edu.eg
X-Sender-IP: 40.107.22.60
Importance: high
MIME-Version: 1.0

Pg1ldGEgAHRCc1lcXVpdj0lQ29udGvudC1ueXB1IiBjb250Z5wPSj0ZXB0Z2h0bWw7IGNoYXJzZXQ9dXRmLTg1PjxkaXYgc3R5bGU91kZPTlQtU0laRTogMTJweDsgRk90VC1GQU1JTfK6ICZxdw9001NlZ291IFVjJnF1b
Y29yYXRpb24tY295b3I6IGluaXRPYw7IGZvbntQdmFyaWZudC1sawdhdrhvyZXN6IG5vcmlhbdsgZm9udC12YXJpYw50LWlhchM6IG5vcmlhbdsgZm9udC1zdHJldGNoO1Bpbmhlcm100yBib3JkZXItaW1hZ2U6IGluaXRPY
<
```


Q8 What is the resolved hostname of the sender's IP address?

mail-am6eur05on2060.outbound.protection.outlook.com

Q9 What corporation owns the sender's IP address?

Microsoft Corporation

The screenshot shows the IPinfo website interface. The browser's address bar displays the URL `ipinfo.io/40.107.22.60?lookup_source=search-bar`. The website's header includes navigation links: Products, Data, Why IPinfo?, Pricing, Resources, Docs, Login, and a Sign up button. A search bar at the top left contains the IP address `40.107.22.60`. On the left sidebar, there are tabs for Summary, Geolocation, Privacy, ASN, Company, and Abuse. The main content area displays a 'Summary' section with the following details:

Summary	
ASN	AS8075 - Microsoft Corporation
Hostname	mail-am6eur05on2060.outbound.protection.outlook.com
Range	40.104.0.0/14
Company	Microsoft Corporation
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@microsoft.com

At the bottom of the page, there is a cookie consent banner with the text: "We use essential cookies to make our site work. With your consent, we may also use non-essential cookies to improve user experience, personalize content, customize advertisements, and analyze website traffic. For these reasons, we may share your site usage data with our social media, advertising, and analytics partners. By clicking 'Accept,' you agree to our website's cookie use as described in our [Cookie Policy](#). You can change your cookie settings at any time by clicking 'Preferences.'" The banner includes buttons for "Preferences" and "Accept". The footer shows the country as "Netherlands" and the time as "9:30 PM".

Q10 What was the result of the SPF check?

Pass

Q11 What is the full SPF record of the sender's domain?

v=spf1 include:spf.protection.outlook.com -all

The screenshot displays the MxToolbox SuperTool interface. The main content area shows the results for a TXT lookup on the domain `social.helwan.edu.eg`. The record found is `"v=spf1 include:spf.protection.outlook.com -all"`. Below this, a test result table indicates that the DNS record was published successfully.

Type	Domain Name	TTL	Record
TXT	SOCIAL.helwan.edu.eg	24 hrs	"v=spf1 include:spf.protection.outlook.com -all"

Test	Result
DNS Record Published	DNS Record found

Reported by freu.eun.eg on 12/10/2025 at 1:24:14 PM (UTC -6). just for you.

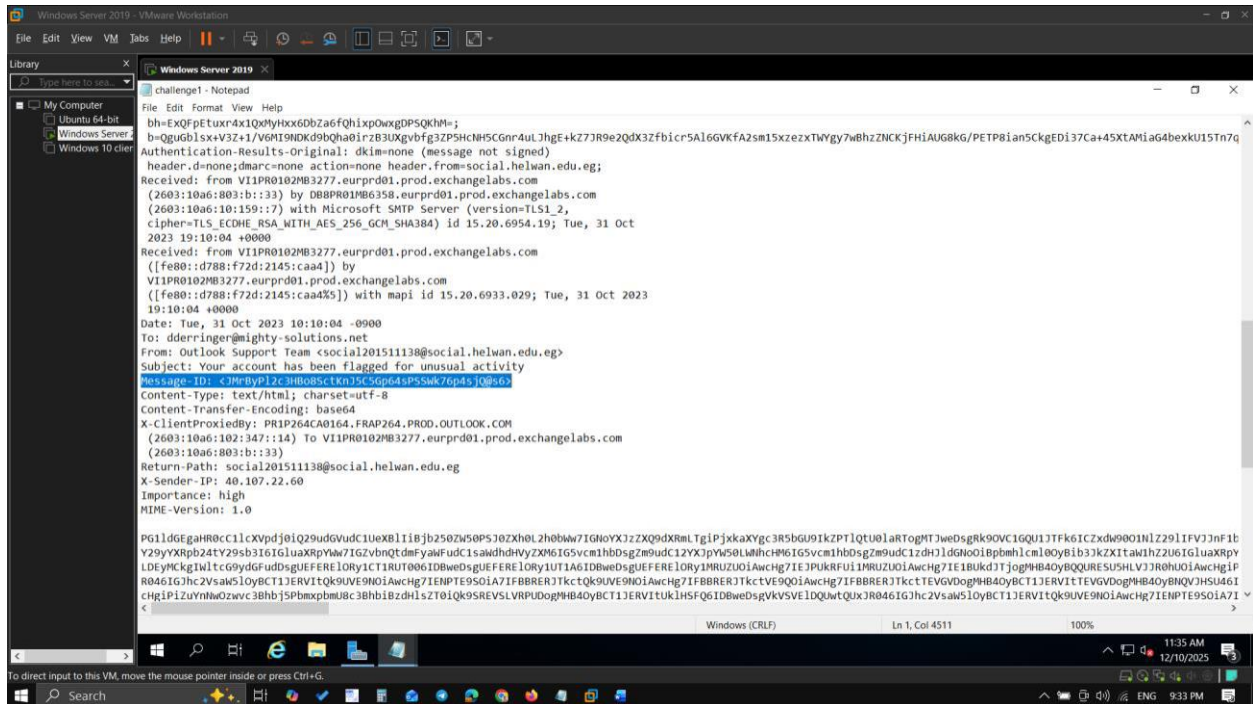
Below the main results, there is a section for the MX record for `mx:social.helwan.edu.eg`. It includes a table with the following data:

Pref	Hostname	IP Address	TTL	Blacklist Check	SMTP Test
0	SOCIAL-helwan-edu-eg.mail.protection.outlook.com	52.101.68.0	24 hrs		

The interface also features a sidebar on the right with various tools like Delivery Center, Inbox Placement, Recipient Complaints, Adaptive Blacklist Monitoring, Mailflow Monitoring, and SPF Flattening.

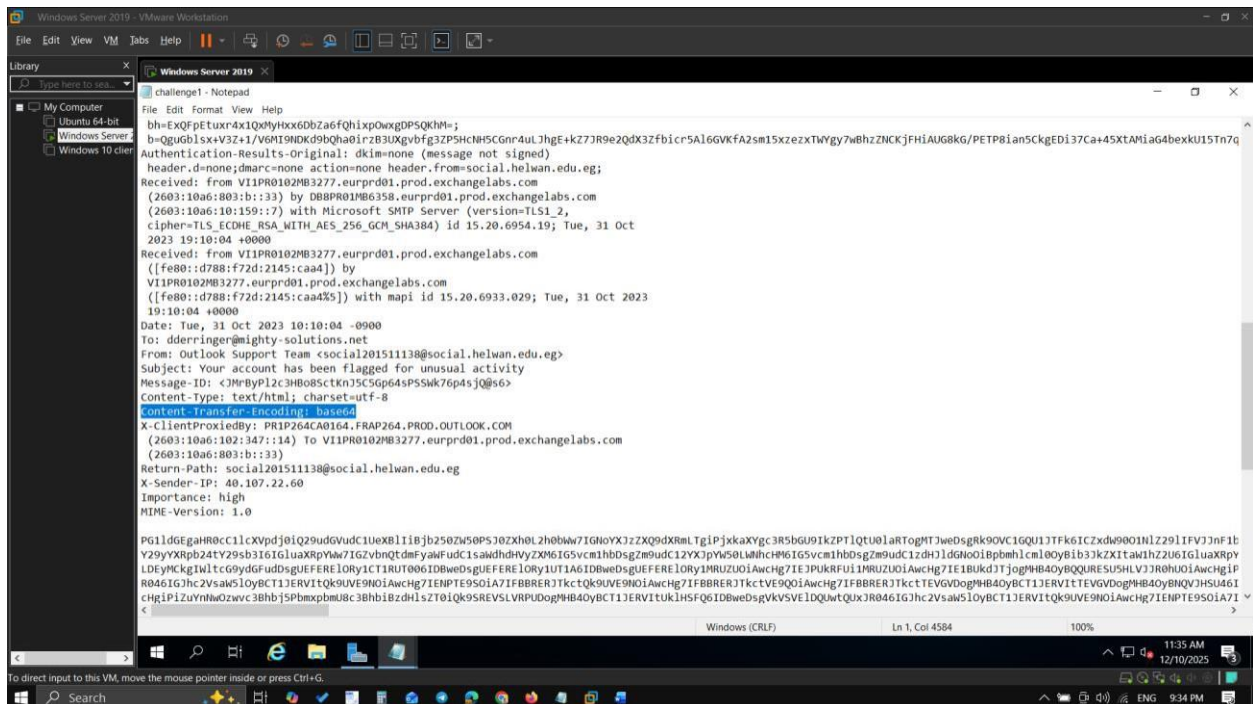
Q12 What is email's Message ID?

JMrByPl2c3HBo8SctKnJ5C5Gp64sPSSWk76p4sjQ@s6

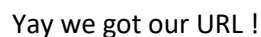


Q13 What type of encoding was used to transfer the email body content?

base64

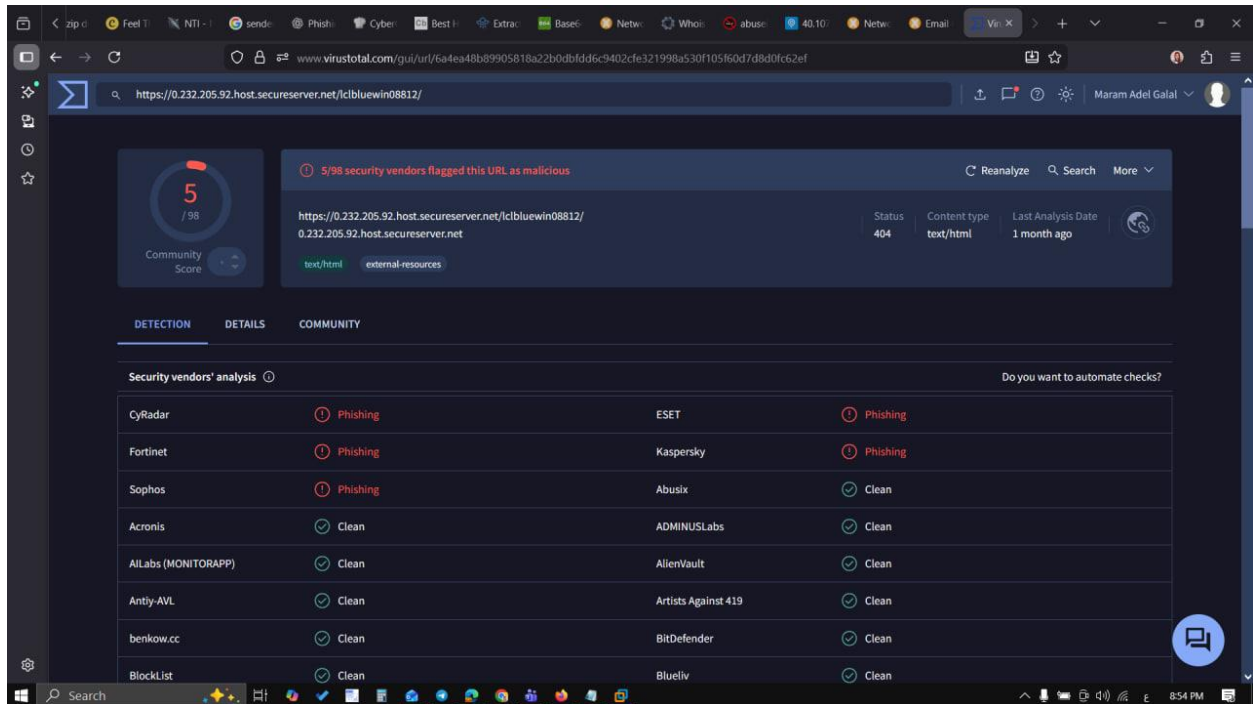


<https://0.232.205.92.host.secureserver.net/lclbluwin08812/>



Q15 Perform a VirusTotal scan on the URL. What verdict did Fortinet assign to it?

Phishing



Q16 [Yes or No] - After your analysis, is this email genuine?

No

Conclusion

- The email impersonates a legitimate brand but uses **external resources hosted on raw.githubusercontent.com**, which is not an official domain of the claimed organization.
- Embedded URLs resolve to an **IP-based or unrelated hosting domain** rather than a trusted, brand-owned domain, which is a strong indicator of phishing.
- The **sender domain does not properly align with the domains used in the links**, suggesting domain spoofing and brand impersonation.
- Email authentication mechanisms (**SPF, DKIM, and DMARC**) are either missing, misaligned, or not clearly tied to the legitimate brand.
- The email is designed to **urge user interaction** (clicking a link or logging in), a common tactic used for credential harvesting.
- Overall structure and infrastructure indicate **malicious intent rather than a legitimate business notification**.

Final Verdict:

Phishing Email – Credential Harvesting / Brand Impersonation